

Derivation of C_{dpmr} and C_n

1 Setup

1.1 Closed-Loop Tracking Error

For a $d=2$ -step measurement delay, the tracking error $\delta x[k] := x_d[k] - x[k]$ in the z -domain is

$$\delta x(z^{-1}) = -H_T(z^{-1}) \delta x_T(z^{-1}) - H_n(z^{-1}) n_x(z^{-1}), \quad (1)$$

with

$$H_T(z^{-1}) = z^{-1} \cdot \frac{1 + (1-\lambda_c)z^{-1} + (1-\lambda_c)z^{-2}}{1 - \lambda_c z^{-1}}, \quad (2)$$

$$H_n(z^{-1}) = z^{-1} \cdot \frac{1-\lambda_c}{1 - \lambda_c z^{-1}}. \quad (3)$$

1.2 Measurement and Residual

$$\delta p_m[k] = \delta x[k-d] + n_x[k], \quad d = 2. \quad (4)$$

$$\delta p_{md}[k] = (1 - a_{pd}) \delta p_{md}[k-1] + a_{pd} \delta p_m[k], \quad H_{\text{LP}}(z^{-1}) = \frac{a_{pd}}{1 - (1 - a_{pd})z^{-1}}. \quad (5)$$

$$\delta p_{mr}[k] = \delta p_m[k] - \delta p_{md}[k], \quad H_{\text{HP}}(z^{-1}) = 1 - H_{\text{LP}}(z^{-1}) = \frac{(1 - a_{pd})(1 - z^{-1})}{1 - (1 - a_{pd})z^{-1}}. \quad (6)$$

1.3 Variance Estimate

$$\hat{\sigma}^2[k] = (1 - a_{cov}) \hat{\sigma}^2[k-1] + a_{cov} \delta p_{mr}^2[k], \quad H_E(z^{-1}) = \frac{a_{cov}}{1 - (1 - a_{cov})z^{-1}}. \quad (7)$$

2 Channel Decomposition of δp_{mr}

Substituting (4) into (6) and using (1),

$$\delta p_{mr}(z^{-1}) = -F_T(z^{-1}) \delta x_T(z^{-1}) + F_N(z^{-1}) n_x(z^{-1}), \quad (8)$$

$$F_T(z^{-1}) = H_{\text{HP}}(z^{-1}) z^{-d} H_T(z^{-1}), \quad (9)$$

$$F_N(z^{-1}) = H_{\text{HP}}(z^{-1}) [1 - z^{-d} H_n(z^{-1})]. \quad (10)$$

Substituting (3) and factoring for $d = 2$,

$$1 - z^{-d} H_n(z^{-1}) = (1 - z^{-1}) \frac{1 + (1-\lambda_c)z^{-1} + (1-\lambda_c)z^{-2}}{1 - \lambda_c z^{-1}}, \quad (11)$$

$$F_N(z^{-1}) = (1 - z^{-1}) z^3 F_T(z^{-1}). \quad (12)$$

Since δx_T and n_x are independent,

$$\text{Var}(\delta p_{mr}) = C_{\text{dpmr}} \sigma_{\delta x_T}^2 + C_n \sigma_{n_x}^2, \quad (13)$$

$$C_{\text{dpmr}} := \frac{1}{2\pi} \int_{-\pi}^{\pi} |F_T(e^{j\omega})|^2 d\omega, \quad C_n := \frac{1}{2\pi} \int_{-\pi}^{\pi} |F_N(e^{j\omega})|^2 d\omega. \quad (14)$$

3 Closed Form for C_{dpmr}

Evaluating (14) by Parseval and residues at $z = 1 - a_{pd}$ and $z = \lambda_c$,

$$C_{\text{dpmr}}(\lambda_c, a_{pd}) = (1 - a_{pd})^2 \left[2 \cdot \frac{(1 - a_{pd})(1 - \lambda_c)}{1 - (1 - a_{pd})\lambda_c} + \frac{2}{2 - a_{pd}} \cdot \frac{1}{(1 + \lambda_c)[1 - (1 - a_{pd})\lambda_c]} \right]. \quad (15)$$

4 Closed Form for C_n

By (12), $|F_N(e^{j\omega})|^2 = 2(1 - \cos \omega)|F_T(e^{j\omega})|^2$, so

$$C_n = 2 C_{\text{dpmr}} - 2 R_{f_T}[1], \quad R_{f_T}[1] := \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \omega \cdot |F_T(e^{j\omega})|^2 d\omega. \quad (16)$$

Evaluating $R_{f_T}[1]$ by the same residue method with one extra factor of z at each pole,

$$C_n(\lambda_c, a_{pd}) = \frac{2(1 - a_{pd})^2}{2 - a_{pd}} \left[1 + \frac{(1 - a_{pd})^2 a_{pd} (1 - \lambda_c)}{1 - (1 - a_{pd})\lambda_c} + \frac{(1 - \lambda_c)^2}{(1 + \lambda_c)[1 - (1 - a_{pd})\lambda_c]} \right]. \quad (17)$$

5 Mean of the Variance Estimate

For any stable LTI filter $H(z^{-1})$ acting on a stationary input $x[k]$ ($E[x[k]] = \mu_x$, $E[x[k]x[k + \tau]] = R_x(\tau)$) and output $y[k] = H \cdot x[k]$,

$$E[y[k]] = H(z=1) \cdot E[x[k]], \quad (18)$$

$$\text{Var}(y[k]) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 S_x(\omega) d\omega. \quad (19)$$

Direct evaluation gives $H_{\text{HP}}(z=1) = 0$ and $H_E(z=1) = a_{\text{cov}}/a_{\text{cov}} = 1$, hence δp_{mr} is zero-mean and $\text{Var}(\delta p_{mr}) = E[\delta p_{mr}^2]$, so by (18),

$$E[\hat{\sigma}^2] = H_E(z=1) \cdot E[\delta p_{mr}^2] = \text{Var}(\delta p_{mr}) = C_{\text{dpmr}} \sigma_{\delta x_T}^2 + C_n \sigma_{n_x}^2, \quad (20)$$

independent of a_{cov} .